

C A Anusha

Experienced **Senior Embedded Engineer** with 8+ years in **real-time, safety-critical systems** for aerospace, automotive, and semiconductor industries. Proven expertise in designing secure, scalable systems and optimizing embedded platforms for performance and reliability.

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EXPERIENCE

SENIOR ENGINEER / Stellantis, Bengaluru

Dec 2023 - Present

SR. SOFTWARE ENGINEER / Lam Research, Bengaluru

June 2021 – Dec 2023

EMBEDDED ENGINEER II / Honeywell, Bengaluru

Jul 2017 – June 2021

PROJECTS

FOTA (Firmware Over-The-Air) Implementation

- Designed a secure firmware update mechanism for Stellantis vehicle ECUs, enabling secure, remote firmware updates with minimal downtime.
- Implemented AES-256 encryption to protect against cyber threats, ensuring the integrity and authenticity of the firmware update process.
- Developed the OTA platform using Yocto-based Embedded Linux, optimizing the deployment process and ensuring the scalability for over-the-air updates.

Secure Boot Implementation

- Designed and implemented Secure Boot for embedded systems to ensure trusted execution and secure device initialization.
- Developed cryptographic mechanisms, including AES-256 encryption, to verify firmware integrity during the boot process, preventing unauthorized code execution.
- Integrated Secure Boot into the bootloader process, ensuring resilience against cyber threats and protection of sensitive data from tampering.

Designed a remote IO communication protocol

- Developed CDAIO, a proprietary high-priority transfer protocol over TCP/IP, optimizing real-time communication between process controllers and remote IO modules.
- Enhanced protocol performance, achieving a 50ms response time with seamless integration of Modbus TCP, EtherNet/IP, and Foundation Fieldbus (FF) for industrial automation.
- Engineered an advanced event-handling mechanism, ensuring low-latency, deterministic communication for safety-critical applications.

Hardware Fine Tuning for Enhanced Efficiency

- Developed algorithms to fine-tune hardware performance, resulting in 76% increased device coverage and enhanced system efficiency.
- Collaborated with hardware teams to integrate software optimizations into semiconductor equipment, ensuring real-time performance improvements and reliability in high-stakes environments.
- Tested and validated the system through simulation environments and real-world conditions to ensure compatibility and optimal performance in manufacturing processes.

AWARDS & RECOGNITION

Rookie of the Quarter (2017, 2024): Awarded for outstanding performance at Honeywell and Stellantis

Go Beyond Award (2018, 2019): Awarded for exceptional contributions

Hackathon Challenge (2018): Led a team to victory at Honeywell hackathon

Be Committed (2020): Awarded for dedication and consistent high performance

Best Hackathon Project (2021): Recognized for the innovative and impactful project developed at Lam Research hackathon.

Intel® oneAPI Hackathon (2023): Secured third place in a global hackathon showcasing innovative solutions using Intel oneAPI

Namma Yatri Hackathon (2023): Honorary mentions Namma Yatri Hackathon

Intel oneAPI Innovator (2023): Recognized by Intel for outstanding contributions and advocacy in oneAPI development

EDUCATION

BACHELOR OF ENGINEERING

Major: Computer Science

Aug 2013 – Jun 2017

Aggregate: 75.4%

GSSSIETW, Mysuru

SKILLS

PROGRAMMING LANGUAGE

C/C++, Python, Java

Bash

RTOS

QNX, VeraOS,

INTEGRITY OS, PSOS

SDK

Yocto, Kepler

PROTOCOLS

TCP/IP, UDP, MQTT, CAN,

EtherNet/IP, Modbus TCP,

Foundation Fieldbus (FF),

PROFINET,

CDA, CDAIO

BOOTP

STANDARDS

IEC 61508

PROJECT MANAGEMENT

VERSION CONTROL

Git, SVN, ClearCase

Jenkins, TeamCity

TOOLS

Eclipse, VSCode, PyCharm, Sublime

Text/Merge

Wireshark, WinSCP, Putty,

FileZilla

OBd, JTAG, GDB

OTHERS

JIRA, Confluence

Wiki. Phabricator

KEY STRENGTHS

Deliberative

Strategic

Analytical

Command

Maximizer

Object Oriented Programming

Linux Specialist

INTERESTS

Participating in Hackathons

Protocol Design

Embedded Systems Design

Cybersecurity for embedded devices

Customer Centric Product Design